## 5.1 $Ax = b,\ A \in \mathbb{R}^{m \times n}$ $b \neq 0$

a) $m < n \to A$ je široká matice

$$\begin{pmatrix} 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} b_1 \end{pmatrix}$$

$b_1 = 0 \cdot x_1 + 0 \cdot x_2 = 0 \to b = 0 \to$ NEPLATÍ

b) $m > n \to A$ je vysoká

$$\begin{pmatrix} a_1 \\ a_2 \end{pmatrix} (x_1) = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}$$

$$\left.\begin{aligned} b_1 &= a_1 x_1 \\ b_2 &= a_2 x_1 \end{aligned}\right\} \mapsto$$ má řešení pro alespoň jedno $a_{i,j} \neq 0 \to$ NEPLATÍ

c) $m < n \to$ široká

A má plnou hodnost $\to \operatorname{rank}(A) = \min(m,n) = m \to$ nezávislé řádky $\to$
$\operatorname{rng}(A) = \mathbb{R}^n \to$ PLATÍ

## 5.3

a) $(a_1, \dots, a_m) \in \mathbb{R}^n \to x = (x_1, \dots, x_n) \in \mathbb{R}^n$, který minimalizuje ECI

$$f(x) = \sum_{i=1}^{m} \|x - a_i\|^2$$

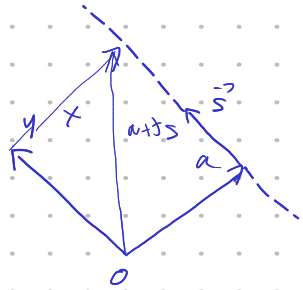
$$\min \sum_{i=1}^{m} \|x - a_i\|^2 \to \min \|Pu - q\|^2$$

$$\sum_{i=1}^{m} \|x - a_i\|^2 = \sum_{i=1}^{m} \left((x_1 - a_{m,1})^2 + \dots + (x_n - a_{m,n})^2\right) = \left\| \begin{pmatrix} x - a_1 \\ \vdots \\ x - a_m \end{pmatrix} \right\|^2 = \left\| \begin{pmatrix} I_1 \\ \vdots \\ I_m \end{pmatrix} x - \begin{pmatrix} a_1 \\ \vdots \\ a_m \end{pmatrix} \right\|^2$$

$$P = \begin{pmatrix} I_1 \\ \vdots \\ I_m \end{pmatrix} \qquad u = x \qquad q = \begin{pmatrix} a_1 \\ \vdots \\ a_m \end{pmatrix}$$

5.3

b)




$x+y=a+ts \rightarrow x=a+ts-y$

$\min_x \|x+y-a-t\cdot s\| \rightarrow P=\begin{pmatrix} I_1 \\ \vdots \\ I_n \end{pmatrix}, u=x, q=y-a-t\cdot s$

5.8

$u=\begin{pmatrix}2\\1\\-3\end{pmatrix} \quad v=\begin{pmatrix}1\\-1\\1\end{pmatrix} \longrightarrow x=\begin{pmatrix}2\\0\\1\end{pmatrix}$

c)

$A=\begin{pmatrix}2 & 1\\1 & -1\\-3 & 1\end{pmatrix} \rightarrow Ax=b \rightarrow B=(A|x)$

$A^T B=\begin{pmatrix}2 & 1 & -3\\1 & -1 & 1\end{pmatrix}\left(\begin{array}{cc|c}2 & 1 & 2\\1 & -1 & 0\\-3 & 1 & 1\end{array}\right)=\left(\begin{array}{cc|c}4+1+9=14 & 2-1-3=-2 & 4+0-3=1\\2-1-3=-2 & 1+1+1=3 & 2-0+1=3\end{array}\right) \rightarrow$

$\rightarrow \left(\begin{array}{cc|c}14 & -2 & 1\\-2 & 3 & 3\end{array}\right) \sim \left(\begin{array}{cc|c}14 & -2 & 1\\0 & 19 & 22\end{array}\right) \begin{array}{l}R_1\\7R_2+R_1\end{array}$

$P_2=\frac{22}{19}$

$14P_1=2P_2+1$

$14P_1=\frac{44}{19}+1=\frac{63}{19}$

$P_1=\frac{63}{19}\cdot\frac{1}{14}=\frac{7\cdot 9}{19(7\cdot 2)}=\frac{9}{19\cdot 2}=\frac{9}{38}$

$\text{proj}(x)=\frac{9}{38}u+\frac{44}{38}v=\frac{1}{38}\begin{pmatrix}18+44\\9-44\\-27+44\end{pmatrix}=\frac{1}{38}\begin{pmatrix}62\\-35\\17\end{pmatrix}$

d)

$\begin{pmatrix}2\\0\\1\end{pmatrix}-\begin{pmatrix}62\\-35\\17\end{pmatrix}\cdot\frac{1}{38}=\frac{1}{38}\begin{pmatrix}76-62\\0+35\\38-17\end{pmatrix}=\frac{1}{38}\begin{pmatrix}14\\35\\21\end{pmatrix}$

5.17

$$\left\| \begin{bmatrix} a \\ b \end{bmatrix} \right\|^2 = \|a\|^2 + \|b\|^2 \quad \longrightarrow \quad c = \begin{bmatrix} a \\ b \end{bmatrix}$$

$$\left( \sqrt{\sum_{i=1}^{2n} |c_i|^2} \right)^2 = \left( \sqrt{\sum_{i=1}^{n} a_i^2} \right)^2 + \left( \sqrt{\sum_{i=1}^{n} b_i^2} \right)^2$$

$$\sum_{i=1}^{n} c_i^2 + \sum_{i=n+1}^{2n} c_i^2 = \sum_{i=1}^{n} a_i^2 + \sum_{i=1}^{n} b_i^2$$

$$\uparrow$$

$c_1 = a_1$

$\vdots$

$c_n = a_n$

$c_{n+1} = b_1$

$c_{2n} = b_n$

$$\sum_{i=1}^{n} a_i^2 + \sum_{i=1}^{n} b_i^2 = \sum_{i=1}^{n} a_i^2 + \sum_{i=1}^{n} b_i^2 \qquad \square$$